SPSS Data Analysis Project

College Life Satisfaction Survey Analysis

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Introduction

The results of the College Life Satisfaction (CLS) survey provide an insightful look at how students' satisfaction varies over time with respect to many aspects of their college experience. This project aims to evaluate these changes and correlations using statistical analyses conducted with SPSS. In addition to CLS scores for 2016 and 2017, the dataset includes factors such as gender distribution, academic year, commuter status, and on-campus living status. The project focuses on how student happiness varies and what factors might influence these changes, examining whether there are significant differences or predictive links within and across the dataset.

Descriptive Statistics

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation	Variance
College life satisfaction score for the year 2016. 0 – least satisfied; 100 – most satisfied	66	33.00	99.00	72.2424	17.37732	301.971
College life satisfaction score for the year 2017. 0 – least satisfied; 100 – most satisfied	66	49.00	100.00	79.4697	13.03394	169.884
Valid N (listwise)	66					

The above image shows the descriptive statistics on several areas of college life happiness across two years, 2016 and 2017. The College Life Satisfaction score for 2016 (CLS2016) runs from 0 to 100, with an average satisfaction score of 72.24. The standard deviation for this score is rather high, at 17.38, indicating that the participants' replies were fairly diverse. The minimum recorded score is 33, and the greatest is 99, indicating that some students were very satisfied while others were not. In comparison, the College Life Satisfaction score for 2017 (CLS2017) shows a significant improvement, with a mean score of 79. 47. The standard deviation for 2017 is lower at 13.03, indicating that responses were more consistent than in 2016. The least 2017 score is 49, and the maximum is 100, indicating that respondents' satisfaction levels have increased generally.

Gender of the Respondents

gender of the respondents

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	male	31	47.0	47.0	47.0
	female	35	53.0	53.0	100.0
	Total	66	100.0	100.0	

The above data is described in two variables, male and female and the gender distribution among respondents is well equal, with 47% identifying as male and 53% as female respondents. This fairly balanced representation means that gender differences in college life satisfaction can be investigated without considerable bias.

College Year

College Year

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Senior	22	33.3	33.3	33.3
	Junior	23	34.8	34.8	68.2
	Sophomore	21	31.8	31.8	100.0
	Total	66	100.0	100.0	

The proportion of responses by year of college is likewise very even. Juniors account for the largest group (34.8%), followed by seniors (33.3%) and sophomores (31.8%). This distribution implies that the dataset accurately represents students at various phases of their college education, which can provide insight into how satisfaction levels may fluctuate as students proceed through their academic careers.

Do you: Commute/Live on campus

Do you: Commute / Live on campus

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Commute	44	66.7	66.7	66.7
	Live on Campus	22	33.3	33.3	100.0
	Total	66	100.0	100.0	

The data on commuting versus living on campus shows that the majority of respondents (66.7%) commute to campus, while 33.3% reside there. This distinction could be a crucial component in determining happiness levels, as commuting and living on campus may provide different experiences and difficulties to students. For example, students who reside on campus may have greater possibilities to participate in university events, which could boost their satisfaction levels.

1. Does the college year impact whether WSU students commute/live on campus?

The null hypothesis (H0): There is no association between the college year and students' commuting or living on campus status. (Whether students commute or live on campus is unaffected by their college year.)

Alternate Hypothesis (H1): There is a link between the college year and students' commuting or living on campus status. (College year influences whether students commute or live on campus.)

The Chi-Square Test of Independence is the proper test statistic for this research question. This test will allow us to see if there is a statistically significant link between the two categorical variables "College Year" and "Commute/Live on Campus."

College Year * Do you: Commute / Live on campus Crosstabulation

			Do you: Commute / Live on campus		
			Commute	Live on Campus	Total
College Year	Senior	Count	17	5	22
		% within College Year	77.3%	22.7%	100.0%
		% within Do you: Commute / Live on campus	38.6%	22.7%	33.3%
	Junior	Count	18	5	23
		% within College Year	78.3%	21.7%	100.0%
		% within Do you: Commute / Live on campus	40.9%	22.7%	34.8%
	Sophomore	Count	9	12	21
		% within College Year	42.9%	57.1%	100.0%
		% within Do you: Commute / Live on campus	20.5%	54.5%	31.8%
Total		Count	44	22	66
		% within College Year	66.7%	33.3%	100.0%
		% within Do you: Commute / Live on campus	100.0%	100.0%	100.0%

The crosstabulation result shows the distribution of students who commute versus living on campus, organized by college year (Senior, Junior, Sophomore). For instance, 77.3% of seniors commute and 22.7% live on campus. 42.9% of sophomores commute, with 57.1% living on campus.

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	7.862[a]	2	.020
Likelihood Ratio	7.671	2	.022
Linear-by-Linear Association	5.545	1	.019
N of Valid Cases	66		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 7.00.

The Pearson Chi-Square score obtained from the test was 7.862 with two degrees of freedom (df). The p-value (asymptotic significance) for this test was 0.020.

The p-value is smaller than the usually recognized significance level of 0.05, thus we may conclude that the result is statistically significant. This suggests that there is enough evidence to reject the null hypothesis, which proposed no link between college year and commuting/living on campus.

Based on the results of the Chi-Square test, we conclude that there is a statistically significant relationship between students' college years and whether they commute or live on campus. Seniors and juniors are more likely to commute, whereas sophomores are more likely to reside on-campus. The findings indicate that the chance of living on campus diminishes as students advance through their college years.

2. Is the mean CLS score for the year 2016 different than 65?

Null Hypothesis (H_0): The average College Life Satisfaction (CLS) score for 2016 is 65.

Alternative Hypothesis (H_1): The average College Life Satisfaction (CLS) score in 2016 differs from 65.

A one-sample t-test is used to see if the mean CLS score for 2016 differed from 65. The t-test is the proper test statistic for this hypothesis.

One-Sample Statistics

	N	Mean	Std. Deviation	Std. Error Mean
College life satisfaction score for the year 2016. 0 – least satisfied; 100 – most satisfied	66	72.2424	17.37732	2.13900

One-Sample Test

	Test Value = 65						
			Significance			95% Confidence Interval of the Difference	
	t	df	One-Sided p	Two-Sided p	Mean Difference	Lower	Upper
College life satisfaction score for the year 2016. 0 – least satisfied; 100 – most satisfied	3.386	65	<.001	.001	7.24242	2.9705	11.5143

One-Sample Effect Sizes

				95% Confidence Interval	
		Standardizer[a]	Point Estimate	Lower	Upper
College life satisfaction score for the year 2016. 0 – least satisfied; 100 – most satisfied	Cohen's d	17.37732	.417	.164	.667
	Hedges' correction	17.58109	.412	.162	.659

a. The denominator used in estimating the effect sizes.
Cohen's d uses the sample standard deviation.
Hedges' correction uses the sample standard deviation, plus a correction factor.

In 2016, the mean CLS score was 72.24 (SD = 17.38), which differed considerably from the test value of 65 ($t(65) = 3.39$, $p < 0.001$). The 95% confidence interval for the mean difference was 2.97 to 11.51, which did not contain zero. This indicates that the null hypothesis has been rejected, and we can conclude that the average College Life Satisfaction score in 2016 is statistically substantially higher than 65.

Furthermore, the effect size determined with Cohen's d is 0.417, indicating a small-to-medium impact size. This shows that the gap between the CLS score and the test result of 65 is visible, but not significant.

3. Do males have different CLS score for year 2017 than females?

The null hypothesis (H_0): Males and females have the same College Life Satisfaction (CLS) scores in 2017.

Alternate Hypothesis (H_1): Male and female students have significantly different College Life Satisfaction (CLS) scores in 2017.

The independent Samples t-test is the most appropriate statistical test for comparing the means of two separate groups (male and female). This test examines the average scores of two distinct groups to determine whether there is a statistically significant difference between them.

Group Statistics

	gender of the respondents	N	Mean	Std. Deviation	Std. Error Mean
College life satisfaction score for the year 2017. 0 – least satisfied; 100 – most satisfied	male	31	82.9032	13.61948	2.44613
	female	35	76.4286	11.86762	2.00599

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means							
						Significance				95% Confidence Interval of the Difference	
		F	Sig.	t	df	One-Sided p	Two-Sided p	Mean Difference	Std. Error Difference	Lower	Upper
College life satisfaction score for the year 2017. 0 – least satisfied; 100 – most satisfied	Equal variances assumed	.769	.384	2.064	64	.022	.043	6.47465	3.13694	.20790	12.74141
	Equal variances not assumed			2.047	59.982	.023	.045	6.47465	3.16347	.14673	12.80258

Independent Samples Effect Sizes

				95% Confidence Interval	
		Standardizer[a]	Point Estimate	Lower	Upper
College life satisfaction score for the year 2017. 0 – least satisfied; 100 – most satisfied	Cohen's d	12.71888	.509	.016	.998
	Hedges' correction	12.87040	.503	.016	.987
	Glass's delta	11.86762	.546	.041	1.042

a. The denominator used in estimating the effect sizes.
Cohen's d uses the pooled standard deviation.
Hedges' correction uses the pooled standard deviation, plus a correction factor.
Glass's delta uses the sample standard deviation of the control group.

Levene's Test for Equality of Variances: This test determines whether the variances between two groups are equal. If the p-value exceeds 0.05, we assume equal variances. In the output, the p-value for Levene's test is 0.384, suggesting that we assume equal variances.

The t-test for equality of Means: The t-value is 2.064, with 64 degrees of freedom (df). The two-tailed p-value is 0.043, which is less than the threshold of 0.05. This means that we reject the null hypothesis while accepting the alternate hypothesis.

Mean Difference: The mean difference between male and female CLS scores in 2017 is 6.47, with males having a higher mean score (82.9) than females (76.4).

The 95% confidence interval for the difference in means is [0.2079, 12.7141], implying that the true mean difference is within this range.

Based on a p-value of 0.043, which is less than 0.05, we reject the null hypothesis. Male and female respondents had significantly different College Life Satisfaction scores in 2017. Males expressed much higher satisfaction ratings than females.

The effect size (Cohen's d) is 0.509, indicating a moderate effect size, implying that the satisfaction gap between males and females is both statistically significant and practically meaningful.

4. Is the CLS score for the two years (2016 and 2017) different?

Null Hypothesis (H_0): The mean CLS scores in 2016 and 2017 are identical. In other words, there is no discernible difference in scores.

Alternative Hypothesis (H_1): The mean CLS scores in 2016 and 2017 differ.

A paired samples t-test is appropriate for testing the hypotheses because we are comparing two related samples: CLS scores from the same set of students in two separate years (2016 and 2017). The paired t-test determines if the average difference between the paired observations (i.e., the 2016 and 2017 scores) is significantly different from zero.

Paired Samples Statistics

		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	College life satisfaction score for the year 2016. 0 – least satisfied; 100 – most satisfied	72.2424	66	17.37732	2.13900
	College life satisfaction score for the year 2017. 0 – least satisfied; 100 – most satisfied	79.4697	66	13.03394	1.60437

The mean CLS score in 2016 was 72.24 with a standard deviation of 17.38, whereas in 2017, it was 79.47 with a standard deviation of 13.03. The number of students (N) is 66 for both years, hence the analysis is based on 66 pairs of observations.

Paired Samples Correlations

		N	Correlation	Significance	
				One-Sided p	Two-Sided p
Pair 1	College life satisfaction score for the year 2016. 0 – least satisfied; 100 – most satisfied & College life satisfaction score for the year 2017. 0 – least satisfied; 100 – most satisfied	66	-.142	.128	.255

Paired Samples Test

		Paired Differences					t	df	Significance	
					95% Confidence Interval of the Difference					
		Mean	Std. Deviation	Std. Error Mean	Lower	Upper			One-Sided p	Two-Sided p
Pair 1	College life satisfaction score for the year 2016. 0 – least satisfied; 100 – most satisfied - College life satisfaction score for the year 2017. 0 – least satisfied; 100 – most satisfied	-7.22727	23.15552	2.85025	-12.91961	-1.53493	-2.536	65	.007	.014

The average difference between the CLS scores for the two years (2017 minus 2016) is -7.23, suggesting that students reported more satisfaction in 2017 than in 2016. The t-value (test statistic) is -2.536, with 65 degrees of freedom. The two-tailed p-value is 0.014, which is less than the usually accepted significance criterion of 0.05. This indicates that the difference in CLS scores between the two years is statistically significant.

Paired Samples Effect Sizes

					95% Confidence Interval	
			Standardizer[a]	Point Estimate	Lower	Upper
Pair 1	College life satisfaction score for the year 2016. 0 – least satisfied; 100 – most satisfied - College life satisfaction score for the year 2017. 0 – least satisfied; 100 – most satisfied	Cohen's d	23.15552	-.312	-.558	-.064
		Hedges' correction	23.42706	-.309	-.552	-.063

a. The denominator used in estimating the effect sizes.
Cohen's d uses the sample standard deviation of the mean difference.
Hedges' correction uses the sample standard deviation of the mean difference, plus a correction factor.

Cohen's d for this test is -0.312, indicating a small to moderate effect size. This suggests that, while the difference between the years is statistically significant, it is not particularly substantial. The effect size indicates how significant the difference is in practical terms.

The 95% confidence interval for the mean difference is -12.92 to -1.53. This interval excludes 0, providing more evidence that there is a true difference between the two years. In practical terms, this means that we can be 95% certain that the genuine average difference in satisfaction between 2016 and 2017 is within this range and is not attributable to chance.

The data support the alternative hypothesis (H_1). The p-value (0.014) is smaller than the 0.05 significance level, showing a statistically significant difference in CLS scores over the two years. Specifically, average college life satisfaction increased significantly in 2017 compared to 2016. This shows that students' happiness with college life has improved over the past year. Although the effect size is small to moderate, statistical evidence suggests a real and significant rise in student satisfaction from 2016 to 2017.

5. Is the 2016 CLS score for the three college years different? If yes, then explore and do appropriate tests to show where the differences exist. (Bonus 2 points)

Null Hypothesis (H_0): There is no significant variation in the 2016 CLS score between the different college years (Senior, Junior, Sophomore).

Alternative Hypothesis (H_1): The 2016 CLS score varies significantly among college years (Senior, Junior, and Sophomore).

The One-Way ANOVA test statistic is appropriate for this research since we are comparing the CLS score means across three separate groups (Senior, Junior, and Sophomore).

Descriptives

College life satisfaction score for the year 2016. 0 – least satisfied; 100 – most satisfied

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
					Lower Bound	Upper Bound		
Senior	22	83.4545	14.78650	3.15249	76.8986	90.0105	47.00	99.00
Junior	23	61.9130	17.39361	3.62682	54.3915	69.4346	33.00	98.00
Sophomore	21	71.8095	12.56431	2.74176	66.0903	77.5287	46.00	95.00
Total	66	72.2424	17.37732	2.13900	67.9705	76.5143	33.00	99.00

According to the descriptive data, seniors reported the highest level of college life satisfaction (mean = 83.45), followed by sophomores (mean = 71.81), and juniors reported the lowest (mean = 61.91). There is a notable distinction in satisfaction ratings, and the 95% confidence intervals show these variances.

ANOVA

College life satisfaction score for the year 2016. 0 – least satisfied; 100 – most satisfied

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	5223.602	2	2611.801	11.423	<.001
Within Groups	14404.519	63	228.643		
Total	19628.121	65			

The ANOVA test findings revealed a statistically significant difference in CLS scores between the three groups, as demonstrated by an F-value of 11.423 and a p-value of less than 0.001. This shows that the mean satisfaction levels differed greatly by year of college.

ANOVA Effect Sizes[a]

		Point Estimate	95% Confidence Interval Lower	Upper
College life satisfaction score for the year 2016. 0 – least satisfied; 100 – most satisfied	Eta-squared	.266	.085	.413
	Epsilon-squared	.243	.056	.394
	Omega-squared Fixed-effect	.240	.056	.390
	Omega-squared Random-effect	.136	.029	.242

a. Eta-squared and Epsilon-squared are estimated based on the fixed-effect model.

The Eta-squared value of 0.266 suggests that differences in college years account for approximately 26.6% of the variance in satisfaction levels, reflecting a modest effect size. This demonstrates that the academic year has a significant impact on satisfaction levels.

Multiple Comparisons

Dependent Variable: College life satisfaction score for the year 2016. 0 – least satisfied; 100 – most satisfied
Tukey HSD

(I) College Year	(J) College Year	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval Lower Bound	Upper Bound
Senior	Junior	21.54150*	4.50931	<.001	10.7177	32.3653
	Sophomore	11.64502*	4.61309	.037	.5721	22.7180
Junior	Senior	-21.54150*	4.50931	<.001	-32.3653	-10.7177
	Sophomore	-9.89648	4.56386	.085	-20.8512	1.0583
Sophomore	Senior	-11.64502*	4.61309	.037	-22.7180	-.5721
	Junior	9.89648	4.56386	.085	-1.0583	20.8512

*. The mean difference is significant at the 0.05 level.

Post-hoc comparisons using Tukey's HSD test revealed significant differences between Senior and Junior students, as well as Senior and Sophomore students. Seniors reported considerably higher satisfaction scores than juniors (mean difference = 21.54, $p < 0.001$) and sophomores (mean difference = 11.64, $p = 0.037$). However, there was no statistically significant difference in CLS ratings between juniors and sophomores (mean difference = 9.89, $p = 0.085$), indicating that these two groups had equal levels of pleasure.

There is sufficient evidence to conclude that the 2016 CLS score differs significantly across the three college years. Seniors have significantly higher CLS ratings than both Juniors and Sophomores, although Juniors and Sophomores do not differ significantly. Thus, the alternative hypothesis is supported.

6. Are the 2016 CLS scores predictive of 2017 CLS scores?

Null Hypothesis (H_0): 2016 CLS scores do not predict 2017 CLS scores (regression coefficient for 2016 CLS scores = 0).

Alternative Hypothesis (H_1): 2016 CLS scores predict 2017 CLS scores (the regression coefficient for 2016 CLS scores is not zero).

To investigate the predictive relationship between College Life Satisfaction (CLS) scores in 2016 and 2017, a simple linear regression analysis is a suitable statistical test. This test assesses the degree to which one independent variable (2016 scores) predicts the another dependent variable (2017 scores) and helps study the relationship between two continuous variables (the scores from two consecutive years).

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.142[a]	.020	.005	13.00228

a. Predictors: (Constant), College life satisfaction score for the year 2016. 0 – least satisfied; 100 – most satisfied

Model Summary (R^2=0.020): This demonstrates that only 2% of the variance in 2017 CLS scores can be explained by 2016 CLS scores. This shows an unstable connection between the two.

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	222.653	1	222.653	1.317	.255[b]
	Residual	10819.787	64	169.059		
	Total	11042.439	65			

a. Dependent Variable: College life satisfaction score for the year 2017. 0 – least satisfied; 100 – most satisfied

b. Predictors: (Constant), College life satisfaction score for the year 2016. 0 – least satisfied; 100 – most satisfied

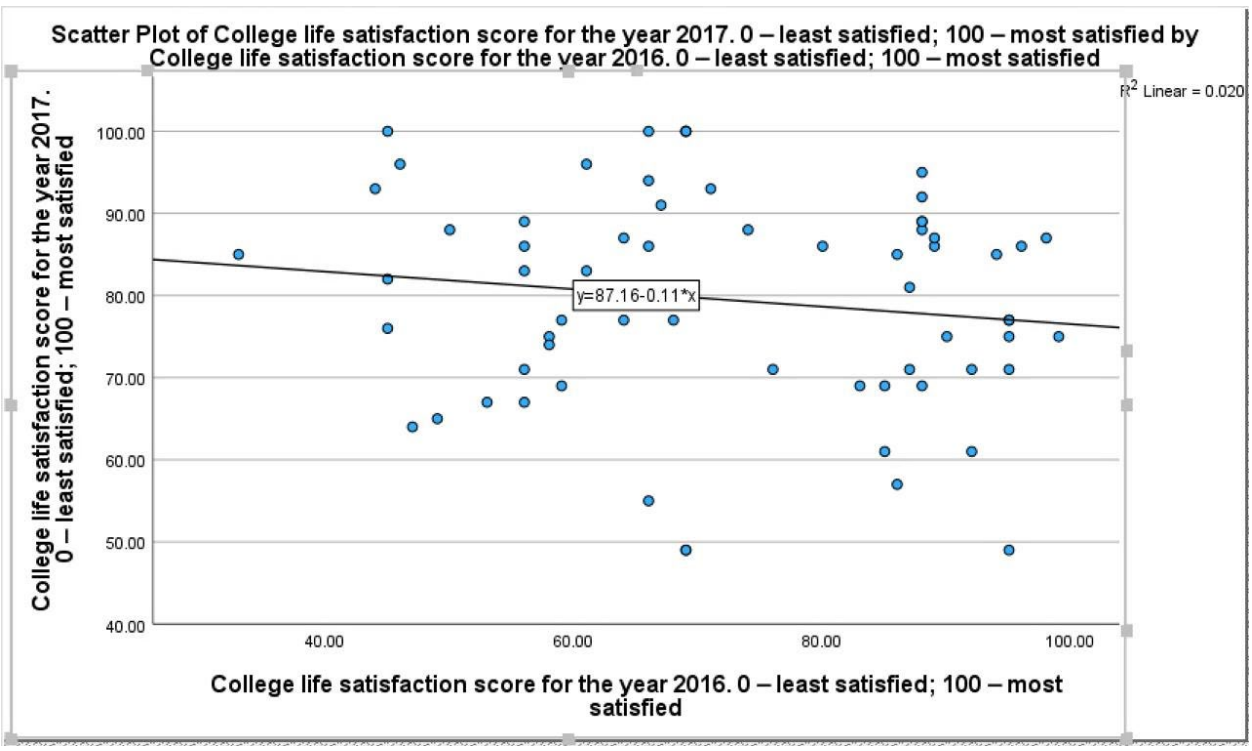
The p-value (0.255) exceeds 0.05, indicating that the model is not statistically significant. This indicates that the 2016 CLS scores do not significantly predict the 2017 CLS scores.

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.
1	(Constant)	87.164	6.893		12.645	<.001
	College life satisfaction score for the year 2016. 0 – least satisfied; 100 – most satisfied	-.107	.093	-.142	-1.148	.255

a. Dependent Variable: College life satisfaction score for the year 2017. 0 – least satisfied; 100 – most satisfied

The 2016 and 2017 scores have a slightly adverse association, according to the regression coefficient (B = -0.107), but the p-value of 0.255 indicates that this effect is not statistically significant.



We are unable to rule out the null hypothesis based on the p-value (0.255) and the confidence interval. This indicates that there is insufficient evidence in the data to support the idea that the CLS scores from 2016 are indicative of the CLS scores from 2017. Stated differently, the two variables in this instance do not have a statistically significant linear connection.

As y = 87.16 - 0.11*x, the scatter plot in the figure displays a negative slope for the regression equation. This means that the 2017 College Life Satisfaction (CLS) score falls

by 0.11 units for every unit rise in the 2016 CLS score. This slight inverse association, however, indicates that the 2016 CLS scores have very little predictive power over the 2017 CLS scores.

Conclusion

Using a variety of statistical factors, the study offers critical insights into how satisfied students are with their college experience. Between 2016 and 2017, there was noticeable variation in the CLS scores, with satisfaction increasing in that year. Gender differences in CLS scores were noted in 2017, with men reporting higher levels of satisfaction than women. The CLS scores from 2016 and 2017 were shown to have little to no statistically significant association, indicating that contentment in one year does not always translate into satisfaction in the following year. The academic year also had a significant impact on the 2016 CLS scores, with seniors expressing higher levels of satisfaction than juniors and sophomores. These results provide valuable information for understanding how college experiences impact students' satisfaction levels and point out areas where colleges may focus their efforts to enhance the overall student experience.